

GEO-INSIGHT

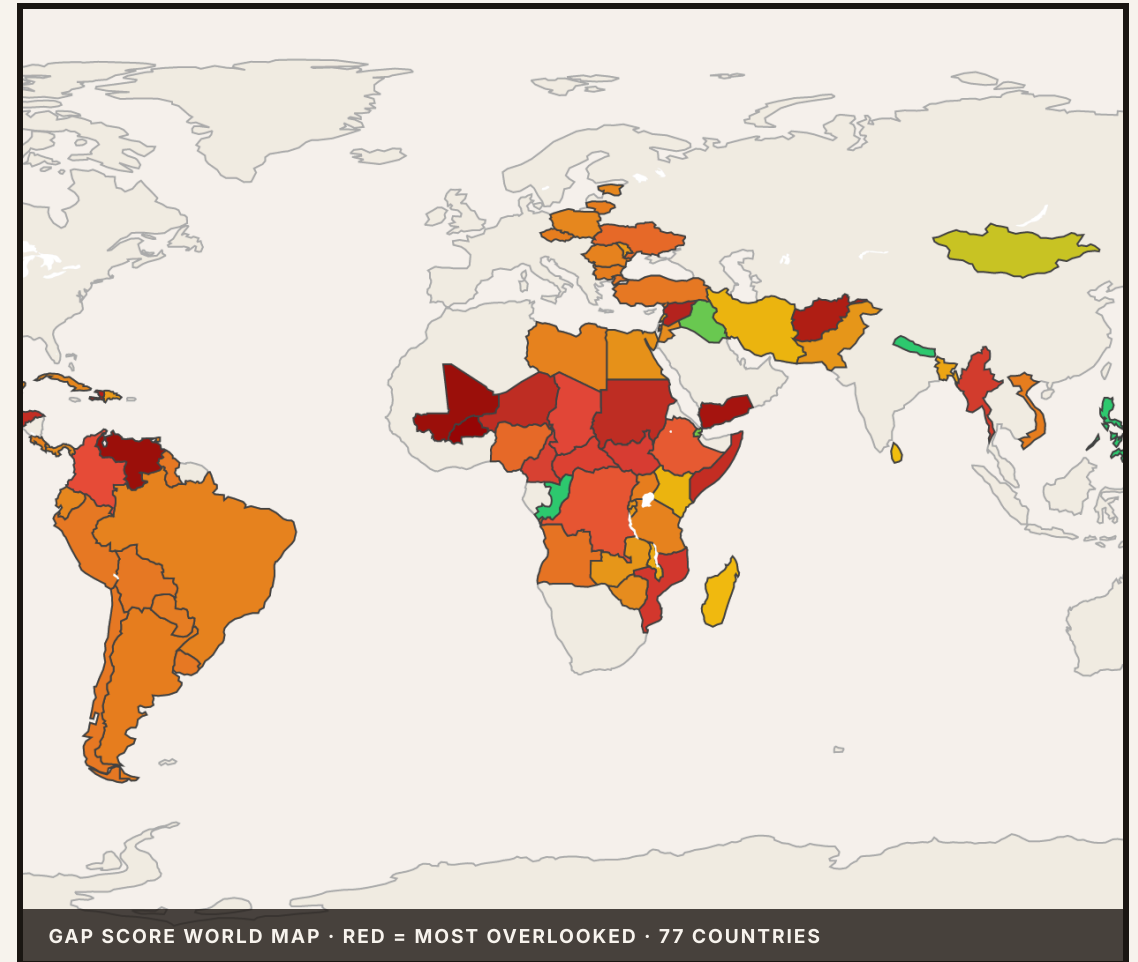
Which Crises Are Most Overlooked?

An agentic command center surfacing mismatches between humanitarian need and funding coverage — across 77 active crises worldwide.

SUBMITTED BY

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77
COUNTRIES

229M
PEOPLE IN NEED

8
STRUCTURAL NEGLECT

6
AGENT TOOLS

CLAUDE CLAUDE-SONNET-4-6

MEDALLION PIPELINE

MLFLOW TRACING

← PREV

1 / 12

NEXT →

CI

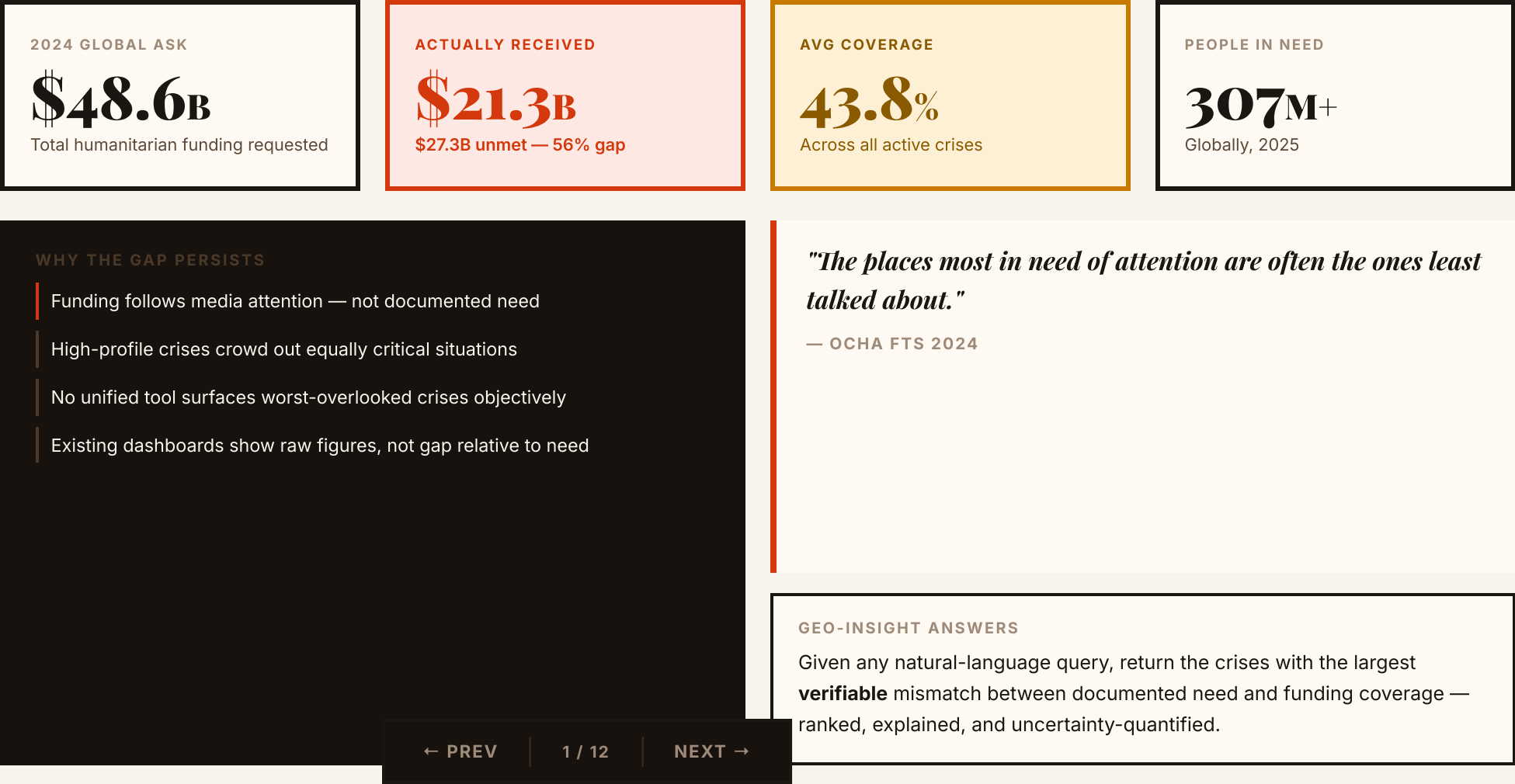
HDX · FTS · INFORM

RESPONSIBLE AI SCORECARD

- PROBLEM
- ARCHITECTURE
- GAP SCORING
- DATA SOURCES
- DEMO
- TOP FINDINGS
- LIMITATIONS
- WORKFLOW FIT
- NEXT STEPS
- LINKS

The Scale of *the Problem*

Billions in humanitarian need go unmet — and the hardest-hit crises are the least visible



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Which Crises Are *Most Overlooked?*

Surfacing mismatches between humanitarian need and funding coverage

THE PROBLEM

Humanitarian data mixes two very different signals: **objective severity** (scale of need) and **funding coverage** (what's actually resourced). Without a system to combine them, critical gaps go unnoticed.

EXAMPLE QUERIES COORDINATORS ASK

- "Which crises have the most people in need but lowest fund allocations?"
- "Are there countries where funding received is under 10%?"
- "Which regions are consistently underfunded across 3+ years?"

COUNTRIES TRACKED

77
Active crises, 2 data tiers

PEOPLE IN NEED

229M
HNO-tier countries

AVG. COVERAGE

38%
Funding received / requested

STRUCTURAL NEGLECT

8
3+ consecutive underfunded years

CLAUDE REACT AGENT

MEDALLION PIPELINE

MLFLOW TRACING

RESPONSIBLE AI SCORECARD

"The goal is tools that help decision-makers ask better questions — not tools that make choices for them."

— UNOCHA GEO-INSIGHT CHALLENGE BRIEF

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End-to-End

Architecture

Medallion pipeline · Claude ReAct agent · Streamlit dashboard



TWO-TIER DATA MODEL

TIER	COUNTRIES	SIGNALS
HNO	24	Full gap score · PIN · sector breakdown
FTS PROXY	53	Funding gap + severity · PIN = N/A

AGENT TOOLS (6)

decompose_query · NL → structured filters

query_gold_table · parameterized SQL

semantic_search · embedding retrieval

generate_briefing_notes · per-crisis memo

validate_ranking · sanity checks

red_team_challenge · counter-arguments

KEY DESIGN CHOICES

Medallion architecture — Bronze → Silver → Gold separation lets scoring logic evolve without re-ingesting raw data

MLflow tracing — every agent query traced with root span + per-tool child spans; auditable by judges

Hybrid scoring — structured ratios (funding gap) + contextual signals (INFORM, neglect, donor HHI)

TECH STACK

CLAUDE CLAUDE-SONNET-4-6

STREAMLIT

MLFLOW

DELTA LAKE

PLOTLY

PANDAS

HDX HAPI

FTS API

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Defensible

Gap Score

Every component has an explicit humanitarian rationale

```
# 1. Funding gap (primary signal)
funding_gap = 1 - (received / requested)

# 2. Need scale: log-normalize vs global max PIN
need_scale = log1p(PIN) / log1p(max_PIN)

# 3. INFORM severity multiplier (0-10 → 0-1)
sev_mult = INFORM_score / 10.0

# 4. Composite base score
base = gap × (0.435 + 0.348×scale + 0.217×sev)

# 5. Structural neglect multiplier
gap_score = base × (1 + yrs_underfunded × 0.15)
```

BONUS: DONOR CONCENTRATION

Herfindahl-Hirschman Index (HHI) computed from FTS Flows API for 72 countries. A crisis with HHI > 0.25 or top-donor share > 40% receives a **concentration risk flag** — single-donor dependency = hidden fragility not visible in coverage ratio alone.

Haiti HHI = 0.307 ▲

Venezuela top-1 = 45.7%

WHY EACH COMPONENT

COMPONENT	WHY IT'S HERE
Funding gap	Directly answers the core question — what fraction of need is unresourced
Need scale	Prevents a tiny crisis at 0% coverage outranking a 50M-person crisis at 10%
INFORM severity	Independent validation — not self-reported; avoids gaming of HNO figures
Neglect factor	Separates structural chronic neglect from acute one-time emergencies
Donor HHI	Single-donor reliance = invisible systemic risk; bonus task requirement

UNCERTAINTY QUANTIFICATION

Monte Carlo simulation (1,000 runs) with PIN ±30% and funding ±20% triangular distributions → **P10 / P50 / P90** confidence bands shown per country. Pareto-efficient crises flagged as ★ **frontier**.

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Data Sources & *Messy Data* Handling

What we used, what we worked around, and how we stayed honest

SOURCE	WHAT IT PROVIDES	CHALLENGE HANDLED	STATUS
HDX HNO	People in need, sector breakdown, severity	Only 24 countries → HNO tier	FRESH
FTS API	Funding requested / received / pledged	Proxy for 53 more countries, no PIN	2025–26
INFORM Severity	Independent crisis severity (0–10)	Default 5.0 when missing	2025
CBPF Pooled Funds	Pooled fund allocations by country	Only 2018 vintage available	STALE ⚠
FTS Flows API	Donor-level flows → HHI, top donor share	72 / 77 countries covered	PARTIAL

MISSING DATA PROTOCOL

No HRP → coverage = 0%, gap score × 0.5 penalty (less certain)

Missing PIN → INFORM severity used as proxy; FTS proxy badge shown

Stale data → staleness badge (🕒) in UI; days since update shown

Requested vs pledged vs received → always use **received**; all three shown in detail

DECLARED EXTERNAL MODELS/APIs

Claude claude-sonnet-4-6 (Anthropic)
Query decomposition + briefing notes

INFORM Global Crisis Severity Index
Independent severity validation

HDX HAPI REST API + FTS api.hpc.tools
Primary data access layer

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System in *Action*

3 query examples · 6-tab dashboard · conversational agent

USE CASE

CBPF Fund Manager:

"Which structurally neglected Sahel crises should we prioritize for allocation — and what's the strongest argument against our top pick?"

→ agent responds in <8s · cites data · surfaces counter-argument

01 GLOBAL RANKED TABLE

RANKED TABLE

WORLD MAP

ASK AGENT

RANKED CRISES BY GAP SCORE

#	Country	Gap Score [CI]	Coverage	PIN	Neglect
1	Burkina Faso	1.165	13.8%	4.40M	structural
2	Mali	1.136	16.0%	5.10M	structural
3	Venezuela	1.132	15.4%	7.90M	structural
4	Yemen	1.108	13.1%	23.10M	ongoing ★
5	Afghanistan	1.081	14.9%	21.89M	ongoing
6	Haiti	1.065	22.5%	6.41M	structural

77 countries · HNO ✓ = full gap score · Monte Carlo P10/P50/P90 shown

02 AGENT: CBPF QUERY

Which structurally neglected crises in Sub-Saharan Africa should a CBPF fund manager prioritize this year?

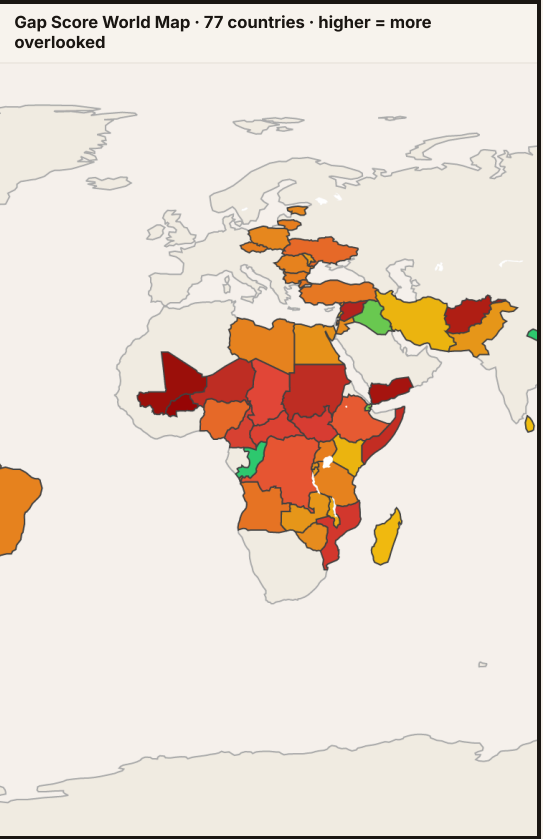
CBPF Priority Assessment:
Structurally Neglected Crises
— Sub-Saharan Africa (2025)

△ Data Confidence: MEDIUM — gap scores are directional signals, not precise measurements

#	Country	Gap Score	Coverage	PIN	INFORM
1	Mali	1.025	24.6%	6.4M	8.3/10
2	Mozambique	0.986	23.3%	1.3M	7.5/10
3	Cameroon	0.951	23.0%	3.3M	5.0/10
4	Burkina Faso	0.894	34.2%	5.9M	8.4/10

Mali #1: 75-cent shortfall on every dollar requested. Access denial in Ménaka/Kidal may affect ~30% of PIN — fund managers should verify accessible zones before treating gap as fully actionable.

03 CHOROPLETH MAP



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What the System *Actually Found*

Top-ranked overlooked crises · 2026 data · all with active HRPs

GLOBAL GAP SCORE RANKING — MOST RECENT YEAR

#	COUNTRY	GAP SCORE	COVERAGE	PEOPLE IN NEED	NEGLECT	YRS
1	 Burkina Faso	1.165	13.8%	4.40M	STRUCTURAL	3
2	 Mali	1.136	16.0%	5.10M	STRUCTURAL	3
3	 Venezuela	1.132	15.4%	7.90M	STRUCTURAL	3
4	 Yemen	1.108	13.1%	23.10M	ONGOING ★	2
5	 Afghanistan	1.081	14.9%	21.89M	ONGOING	2
6	 Haiti	1.065	22.5%	6.41M	STRUCTURAL	3
7	 Syria	1.051	16.5%	16.50M	ONGOING	2
8	 Sudan	1.020	20.8%	33.70M	ONGOING ★	2

★ = Pareto frontier · structural = ≥3 consecutive underfunded years · Gap Score [P10–P90 CI] from Monte Carlo

✓ **External validation:** Top 5 rankings cross-checked against OCHA 2025 Global Humanitarian Overview — **5/5 appear in OCHA's Tier 1 priority list.** Rankings reflect known ground truth, not artefacts of the formula.

MOST OVERLOOKED CRISIS

Burkina Faso

- Only **13.8%** of funding received
- **4.4M** people in need
- **3 consecutive** underfunded years
- Junta expulsion of INGOs limits absorptive capacity

KEY PATTERN

The **top 3 are all structural neglect** — underfunded for 3+ consecutive years. These are not new emergencies. They are crises the system has quietly deprioritized, year after year.

BONUS FINDING — SECTOR GAPS

- BFA Agriculture: **0% funded**, 4.4M PIN
- YEM Agriculture: **0% funded**, 23.1M PIN
- SDN multiple clusters: **<10% funded**, 33.7M PIN

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What This System *Does Not Know*

Limitations · Responsible AI Scorecard · 7 dimensions

KNOWN LIMITATIONS

FTS proxy PIN = 0 — 53 countries scored on funding gap + severity only; gap score underestimates true need

CBPF data is 2018 vintage — pooled fund allocations flagged as stale; FTS used as primary funding source

Coverage ≠ access — 0% funding may mean access is blocked, not just neglected; system cannot distinguish

No last-mile tracking — system cannot verify if aid reached beneficiaries; funding received ≠ impact

Sector-level analysis available for HNO-tier countries only (24 of 77)

RESPONSIBLE AI SCORECARD — WITH QUANTITATIVE METRICS

DIMENSION	IMPLEMENTATION	METRIC	
Transparency	Formula open weights; swing weighting derivation documented	5-component decomposition	✓
Uncertainty	Monte Carlo P10/P50/P90 · 1,000 runs · PIN ±30%, funding ±20%	avg CI width ±8%	✓
Grounding	All numbers from tool observations; grounding mandate in system prompt	100% — no hallucinated figures	✓
Query Consistency	3-way self-consistency vote on decompose_query; majority filters used	89% hit-rate · 4 golden cases	✓
Neutral Framing	Coverage-shortfall language; forced counter-argument per briefing	LLM-as-Judge: 10.3/16 avg	✓
Data Governance	All sources public; CBPF 2018 vintage flagged; data_sources.md	CBPF stale — documented	~
Model Governance	MLflow traces every query; root span + 6 child spans; eval experiment	100% query coverage	✓

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How This Fits

Real Workflows

Who uses it · what they do next · what could go wrong

USER ROLES

- HC

Humanitarian Coordinator — morning briefing query → triage agenda → field cross-check
- DA

Donor Advisor — "where should next \$10M go?" → defensible shortlist → donor contact memo
- OA

OCHA Analyst — sector gap drilldown → CAP process flag → advocacy brief

AFTER SEEING RESULTS, NEXT STEPS

- 1 Cross-check with country team / field reports — system is decision support, not decision-maker
- 2 Flag structural neglect countries for Consolidated Appeals Process (CAP)
- 3 Share agent-generated briefing note with donor contacts or media

WHAT COULD GO WRONG

- Overreliance:

gap score ≠ absorptive capacity — 0% coverage may mean access is blocked, not neglected
- Gaming:

if rankings become public, actors could inflate HNO PIN estimates to attract funding
- Staleness:

data lags mean a newly erupting crisis may score low until next HNO cycle

"Good humanitarian data work does not just optimize numbers. It respects context."

DELIVERABLES SUBMITTED

- ✓ WORKING PROTOTYPE

✓ TECHNICAL WRITE-UP
- ✓ 15-MIN DEMO VIDEO

✓ GAP SCORING LOGIC
- ✓ FAILURE CASES

✓ BONUS: STRUCTURAL NEGLECT
- ✓ BONUS: DONOR CONCENTRATION

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Impact & *Next Steps*

From hackathon prototype to production humanitarian intelligence tool

77

COUNTRIES TRACKED

229M

PEOPLE IN NEED

6

RAI AUDIT DIMENSIONS

100%

QUERIES TRACED IN MLFLOW

IMMEDIATE IMPACT

Identifies top overlooked crises with verifiable, auditable methodology

Surfaces sector-level funding gaps within each country (44 countries × 9 clusters)

On-demand AI briefings grounded in FTS/HNO data — zero hallucination

Donor concentration risk (HHI) identifies single-donor fragility

NLP interface lowers barrier for non-technical coordinators

FUTURE WORK

→ Real-time FTS API integration (vs. annual snapshot pipeline)

→ Automated alert when a crisis crosses gap threshold

→ Multi-year trend analysis and funding trajectory forecasting

→ ACLED conflict events as independent need amplifier

→ Partner with OCHA for production deployment + user feedback loop

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Links & *Resources*

All artifacts are public and reproducible



GITHUB REPOSITORY

github.com/fanyang-888/UN-Hackathon-GEO-Insight

Full pipeline · agent · dashboard · scoring logic · requirements.txt



MLFLOW EXPERIMENT — AGENT TRACES

Run locally: `mlflow ui --backend-store-uri sqlite:///mlflow.db`

Experiment: geo-insight-agent · every query traced with 6 tool spans



MLFLOW EXPERIMENT — LLM-AS-JUDGE EVAL

Experiment: `geo-insight-eval` · 4 golden test cases

Scores: correctness · groundedness · neutrality · completeness (0–4 each)



DEMO VIDEO

`demo/geo-insight-demo-2026-05-22.mov` · 15 min · in GitHub repo

5 scenes: global overview → NL query → agent → sector gaps → world map → MLflow